

Infosys Springboard Virtual Internship 6.0

Completion Report

Batch Number : 6

Start date : 10-11-2025

Name : Harshal Mandliya

Internship Duration : 8 Weeks

1. Project Title

NeuroFleetX: Intelligent Taxi Dispatch System

2. Project Objective

The objective of the NeuroFleetX project was to design and develop a scalable and secure backend system for an intelligent taxi dispatch and fleet management platform. The system aims to efficiently manage users, drivers, and ride requests while integrating AI and machine learning capabilities to enhance transparency, predict estimated arrival times (ETA), and improve overall ride handling. A key goal was to ensure seamless frontend-backend communication and implement robust security mechanisms..

3. Project description in detail

NeuroFleetX is an intelligent taxi dispatch and fleet management system focused on backend development and AI integration. My role in the project was primarily as a **Backend Developer**, where I was responsible for designing, implementing, and securing the complete backend architecture of the system.

The backend was developed using **Spring Boot** with a layered architecture consisting of controller, service, repository, and model layers. This design ensured modularity, scalability, and maintainability. Secure RESTful APIs were developed to handle authentication, user management, driver operations, ride history access, and administrative analytics.

A major contribution was the implementation of **JWT-based authentication and role-based access control (RBAC)**, defining roles such as ADMIN, DRIVER, and USER to ensure secure and authorized access to system resources. Passwords were securely

stored using BCrypt hashing, and proper CORS configurations were applied to prevent unauthorized access.

The project also involved **database design and management using PostgreSQL**, where relational schemas were created using JPA and Hibernate. Core entities such as User, Driver, Ride Request, Taxi (structure only), and ML-related data were modeled to maintain data integrity and consistency.

In addition, I integrated an **existing machine learning ETA prediction model** into the backend through dedicated APIs. The model was used as an external service to predict estimated arrival times based on ride parameters. I was responsible only for integration, not for training or developing the ML model.

Another key feature was the integration of the **OpenAI API** to generate AI-powered, human-readable summaries of ride history. This improved transparency and user experience by converting complex ride data into meaningful explanations. Seamless frontend interoperability was ensured by designing clear API contracts and verifying correct role-based behavior in the UI.

4. Timeline Overview

Week	Activities Completed
Week 1	Requirement analysis and understanding backend scope
Week 2	Backend architecture design and technology stack finalization
Week 3	Database schema design using PostgreSQL and JPA
Week 4	Development of authentication and user management APIs
Week 5	Implementation of JWT security and role-based access control
Week 6	Integration of ML ETA prediction model
Week 7	OpenAI API integration for ride history summarization
Week 8	Testing, debugging, frontend integration, and documentation

5. Project execution details

1. Clone the Repository

The project source code was cloned from the GitHub repository using Git.

2. git clone <<https://github.com/harshalmandliya/NeuroFleetX>>

3. Open the Project

The backend project was opened in an IDE such as **IntelliJ IDEA** or **VS Code** with Java support.

4. **Configure Application Properties**

Database and security configurations were set in the application.properties /file, including

- PostgreSQL database URL, username, and password
- JWT secret and expiration settings
- OpenAI API key for AI-based ride history summarization

5. **Database Setup**

A PostgreSQL database was created, and tables were automatically generated using **JPA & Hibernate** on application startup.

6. **Build the Project**

Maven was used to resolve dependencies and build the application.

7. mvn clean install

8. **Run the Backend Server**

The Spring Boot application was started using:

9. mvn spring-boot:run

or by running the main application class.

10. **API Testing and Verification**

Backend APIs were tested using **Postman** to verify:

- Authentication and authorization
- Role-based access control (ADMIN, DRIVER, USER)
- Secure API responses and data integrity

11. **Frontend Integration**

The backend was connected to the frontend using REST APIs, ensuring smooth data exchange and role-based UI behavior.

6. Challenges Faced

- Integrating an existing ML model without modifying its internal logic
- Ensuring secure role-based access across multiple user types
- Managing data consistency across complex relational entities
- Handling secure API communication between frontend and backend
- Ensuring correct AI-generated explanations for ride history

7. Learnings & Skills Acquired

- ☐ Hands-on experience with **Spring Boot backend development**
- ☐ Strong understanding of **JWT authentication and RBAC security**
- ☐ Practical knowledge of **PostgreSQL, JPA, and Hibernate**

- ☐ Experience in **AI integration using OpenAI API**
- ☐ Understanding of **ML model integration in production systems**
- ☐ Improved skills in REST API design, testing, and debugging

8. Conclusion

The NeuroFleetX project provided valuable hands-on experience in building a secure, scalable, and intelligent backend system. Through this project, I strengthened my backend development skills, gained exposure to AI and ML integration, and learned how to design enterprise-grade systems with strong security practices. The internship significantly enhanced my technical confidence and real-world application development experience.

9. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing the opportunity to work on the NeuroFleetX project. I am thankful to my mentor for continuous guidance and support throughout the internship. I also appreciate my team members for their collaboration and teamwork, which contributed greatly to the successful completion of the project.